MYSQL 10 000

```
▷Run | +Tab | JSON | Select
SELECT d.Nombre, l.Valor, l.Fecha
FROM Dispositivo d
INNER JOIN Lectura l ON d.ID = l.DispositivoID
WHERE d.ID = 2821;
```

6ms

```
▷Run | Select
UPDATE Lectura
SET Estado = 'Revisado'
WHERE Fecha = '2026-03-27';  AffectedRows: 10000  73ms
```

```
▷Run
CREATE INDEX idx_fecha ON Lectura(Fecha);  60ms
```

```
▷Run
DELETE FROM Lectura
WHERE Estado = 'Activo';  10ms
```

```
▷Run | +Tab | JSON
SELECT table_schema AS Base_de_Datos,
       ROUND(SUM(data_length + index_length) / 1024 / 1024, 2) AS Tamaño_MB
FROM information_schema.TABLES
WHERE table_schema = 'db_10000'
GROUP BY table_schema;  11ms
```

db_10000	0.50
Base_de_Datos	db_10000
Tamaño_MB	0.50

MONGO 10 000

```
[2026-03-28 01:20:47] test> use db_10000
[2026-03-28 01:20:48] 1 row retrieved starting from 1 in 365 ms (execution: 45 ms, fetching: 320 ms)
[2026-03-28 01:20:52] db_10000> db.dispositivo.aggregate([
                                { $match: { ID: 2821 } },
                                { $lookup: { from: "lectura", localField: "ID", foreignField: "DispositivoID", as: "historial" } }
                              ]);
[2026-03-28 01:20:52] 1 row retrieved starting from 1 in 395 ms (execution: 74 ms, fetching: 321 ms)
```

```
[2026-03-28 01:20:52] 1 row retrieved starting from 1 in 395 ms (execution: 74 ms, fetching: 321 ms)
[2026-03-28 01:26:22] db_10000> db.lectura.updateMany(
                                  { Fecha: "2026-03-27" },
                                  { $set: { Estado: "Revisado" } }
                                );
[2026-03-28 01:26:22] completed in 212 ms
```

```
[2026-03-28 11:38:03] db_10000> db.lectura.createIndex({ Fecha: 1 });
[2026-03-28 11:38:04] 1 row retrieved starting from 1 in 972 ms (execution: 652 ms, fetching: 320 ms)
```

```
[2026-03-28 11:39:19] db_10000> db.lectura.deleteMany({ Estado: "Activo" });
[2026-03-28 11:39:19] 1 row retrieved starting from 1 in 474 ms (execution: 152 ms, fetching: 322 ms)
```

```
[2026-03-28 11:41:27] db_10000> db.stats();
[2026-03-28 11:41:27] 1 row retrieved starting from 1 in 453 ms (execution: 130 ms, fetching: 323 ms)
```

0.77 MB (802816 / 1048576). Como referencia, en MySQL ocupó 0.50 MB.

MYSQL 1 000 000

```
[2026-03-28 11:52:35] db_1000000> SELECT d.Nombre, l.Valor, l.Fecha
                                  FROM Dispositivo d
                                  INNER JOIN Lectura l ON d.ID = l.DispositivoID
                                  WHERE d.ID = 69718
[2026-03-28 11:52:36] 0 rows retrieved in 324 ms (execution: 9 ms, fetching: 315 ms)
```

```
[2026-03-28 11:53:42] db_1000000> UPDATE Lectura
                                  SET Estado = 'Revisado'
                                  WHERE Fecha = '2026-03-27'
[2026-03-28 11:53:47] 1,000,000 rows affected in 4 s 282 ms
```

```
[2026-03-28 11:54:35] db_1000000> CREATE INDEX idx_fecha ON Lectura(Fecha)
[2026-03-28 11:54:36] completed in 567 ms
```

```
[2026-03-28 11:55:15] db_1000000> DELETE FROM Lectura
                                  WHERE Estado = 'Activo'
[2026-03-28 11:55:15] completed in 261 ms
```

Base_de_Datos	Tamaño_MB
db_1000000	61.11

MONGODB 1000 000

```
[2026-03-28 11:59:43] db_1000000> db.dispositivo.aggregate([
                                    { $match: { ID: 69718 } },
                                    { $lookup: { from: "lectura", localField: "ID", foreignField: "DispositivoID", as: "historial" } }
                                  ]);
[2026-03-28 11:59:44] 1 row retrieved starting from 1 in 636 ms (execution: 76 ms, fetching: 560 ms)
```

```
[2026-03-28 12:00:54] db_1000000> db.lectura.updateMany(
                                    { Fecha: "2026-03-27" },
                                    { $set: { Estado: "Revisado" } }
                                  );
[2026-03-28 12:00:58] completed in 3 s 994 ms
```

```
[2026-03-28 12:03:07] db_1000000> db.lectura.createIndex({ Fecha: 1 });
[2026-03-28 12:03:08] 1 row retrieved starting from 1 in 1 s 241 ms (execution: 921 ms, fetching: 320 ms)
```

```
[2026-03-28 12:03:54] db_1000000> db.lectura.deleteMany({ Estado: "Activo" });
[2026-03-28 12:03:54] 1 row retrieved starting from 1 in 792 ms (execution: 472 ms, fetching: 320 ms)
```

```
[2026-03-28 12:04:44] db_1000000> db.stats();
[2026-03-28 12:04:44] 1 row retrieved starting from 1 in 423 ms (execution: 104 ms, fetching: 319 ms)
```

avgObjSize,collections,dataSize,db,fsTotalSize,fsUsedSize,indexSize,indexes,objects,ok,scaleFactor,storageSize,totalSize,views

92.8543515,2,185708703,db_1000000,485473984512,51605868544,26165248,3,2000000,1,1,82407424,108572672,0

SIZE: 78.59 MB

MYSQL 100 000 000

```
[2026-03-28 18:33:12] db_100000000> SELECT d.Nombre, l.Valor, l.Fecha
                                    FROM Dispositivo d
                                    INNER JOIN Lectura l ON d.ID = l.DispositivoID
                                    WHERE d.ID = 16649998
[2026-03-28 18:33:12] 0 rows retrieved in 389 ms (execution: 18 ms, fetching: 371 ms)
```

```
[2026-03-28 18:35:05] db_100000000> UPDATE Lectura
                                    SET Estado = 'Revisado'
                                    WHERE Fecha = '2026-03-27'
[2026-03-28 18:41:49] 100,000,000 rows affected in 6 m 44 s 565 ms
```

```
[2026-03-28 18:42:49] db_100000000> CREATE INDEX idx_fecha ON Lectura(Fecha)
[2026-03-28 18:43:35] completed in 45 s 918 ms
```

```
[2026-03-28 18:50:56] db_100000000> DELETE FROM Lectura
                                    WHERE Estado = 'Activo'
[2026-03-28 18:51:19] completed in 23 s 166 ms
```

	Base_de_Datos	Tamaño_MB
1	db_100000000	5958.00

MONGODB 100 000 000

```
[2026-03-28 19:03:07] db_100000000> db.dispositivo.aggregate([
                                      { $match: { ID: 16649998 } },
                                      { $lookup: { from: "lectura", localField: "ID", foreignField: "DispositivoID", as: "his
                                    ]);
[2026-03-28 19:03:49] 1 row retrieved starting from 1 in 42 s 163 ms (execution: 140 ms, fetching: 42 s 23 ms)
```

```
[2026-03-28 19:04:25] db_100000000> db.lectura.updateMany(
                                      { Fecha: "2026-03-27" },
                                      { $set: { Estado: "Revisado" } }
                                    );
[2026-03-28 19:11:09] completed in 6 m 43 s 857 ms
```

```
[2026-03-28 19:28:12] db_100000000> db.lectura.createIndex({ Fecha: 1 });
[2026-03-28 19:29:46] 1 row retrieved starting from 1 in 1 m 33 s 605 ms (execution: 1 m 33 s 227 ms, fetching: 378 ms)
```

```
[2026-03-28 19:32:40] db_100000000> db.lectura.deleteMany({ Estado: "Activo" });
[2026-03-28 19:33:20] 1 row retrieved starting from 1 in 39 s 741 ms (execution: 39 s 391 ms, fetching: 350 ms)
```

avgObjSize,collections,dataSize,db,fsTotalSize,fsUsedSize,indexSize,indexes,objects,ok,scaleFactor,storageSize,totalSize,views
93.854453755,2,18770890751,db_100000000,485473984512,62706962432,2743693312,3,200000000,1,1,5749186560,8492879872,0

Name	Container ID	Image	Port(s)	CPU (%)
basesdedatos	-	-	-	253.75%
mongodb-1	b30158595264	mongo:latest	27017:27017	205.54%
mysql-1	8a883640fdc4	mysql:latest	3306:3306	48.21%

Operación	MySQL (1 Millón)	MongoDB (1 Millón)	MySQL (100 Millones)	MongoDB (100 Millones)
Cruce de datos (JOIN / Lookup)	9 ms	76 ms	18 ms	140 ms
Actualización masiva (UPDATE)	4.28 s	3.99 s	6 min 44 s	6 min 43 s
Creación de Índice	567 ms	921 ms	45.91 s	1 min 33 s
Borrado (Escaneo completo)	261 ms	472 ms	23.16 s	39.39 s
Espacio en Disco	61.11 MB	78.59 MB	5958.00 MB	5482.85 MB

EXTRA

```
[2026-03-28 19:57:12] db_100000000> SELECT DispositivoID, COUNT(*) as TotalLecturas, AVG(Valor) as Promedio
                                    FROM Lectura
                                    GROUP BY DispositivoID
[2026-03-28 23:02:27] 500 rows retrieved starting from 1 in 3 h 5 m 14 s 120 ms (execution: 3 h 5 m 13 s 745 ms, fetching: 375 ms)
```

```
[2026-03-28 20:00:51] db_100000000> db.lectura.aggregate([
                                      { $group: { _id: "$DispositivoID", TotalLecturas: { $sum: 1 }, Promedio: { $avg: "$Valor" } } }
                                    ]);
[2026-03-28 21:53:20] 20 rows retrieved starting from 1 in 1 h 52 m 29 s 252 ms (execution: 241 ms, fetching: 1 h 52 m 29 s 11 ms)
```

```
[2026-03-28 23:56:28] db_100000000> SELECT d.Nombre, AVG(l.Valor) as Promedio
                                    FROM Dispositivo d
                                    INNER JOIN Lectura l ON d.ID = l.DispositivoID
                                    GROUP BY d.ID, d.Nombre
                                    ORDER BY Promedio DESC
                                    LIMIT 5
[2026-03-28 23:56:28] 0 rows retrieved in 336 ms (execution: 6 ms, fetching: 330 ms)
```

```
db.dispositivo.aggregate([  13 h 26 m 53 s
  { $lookup: { from: "lectura", localField: "ID", foreignField: "DispositivoID", as: "historial" } },
  { $unwind: "$historial" },
  { $group: { _id: "$Nombre", Promedio: { $avg: "$historial.Valor" } } },
  { $sort: { Promedio: -1 } },
  { $limit: 5 }
], { allowDiskUse: true });
```